

Laws of Indices — Paper A

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is small enough to handle on paper.
- The laws you need: $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{m \times n}$ $a^0 = 1$
- The multiplication and division laws only work when the **bases are the same**.
- Where a question says “as a single power”, leave your answer in index form. Where it asks for a value, work it out.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – WHAT THE INDEX IS SAYING

1. Complete the table. Each empty box is worth one mark. [6]

Power	Written out in full	Value
2^5		
3^3		
10^4		

The middle column is the whole topic in one box: the index counts how many of the base are multiplied together.

2. Write each of these in index form.

(a) $3 \times 3 \times 3 \times 3 \times 3$ [1]

(b) $2 \times 2 \times 2 \times 7 \times 7$ [1]

3. Work out the value of each power.

(a) 5^3 [1]

(b) 2^7

[1]

SECTION 2 – MULTIPLYING AND DIVIDING

4. Write each answer as a single power.

(a) $2^4 \times 2^3$

[1]

(b) $7^2 \times 7^5$

[1]

5. Write each answer as a single power.

(a) $6^8 \div 6^5$

[1]

(b) $10^7 \div 10^4$

[1]

6. Simplify, leaving each answer as a single power.

(a) $a^6 \times a^2$

[1]

(b) $b^7 \div b^3$

[1]

You are never told what a and b stand for, and you never need to know.

7. Simplify, leaving each answer as a single power.

(a) $c^4 \times c^5$

[1]

(b) $d^{10} \div d^4$

[1]

(c) $e^3 \times e \times e^2$

[1]

SECTION 3 – THINKING ABOUT THE LAWS

8. Explain why $2^3 \times 5^2$ **cannot** be written as a single power, and work out its value. [2]

9. Which is larger, 3^4 or 4^3 ? Show your working. [2]

Swapping the base and the index changes the answer. That is worth seeing once, in numbers.

10. Simplify $2^3 \times 2^4 \div 2^5$.

- (a) Deal with the multiplication first, and write the result as a single power. [1]

- (b) Now apply the division law. [1]

- (c) Write your final answer as an ordinary number. [1]

11. A student writes " $2^3 \times 2^5 = 4^8$ ". Explain the mistake, and give the correct answer as a single power. [2]

12. Write 64 as a power of 2, and then as a power of 4.

[2]

The same number can be written as a power in more than one way. Both answers here are correct.

End of paper. Check every answer has a unit or a form the question actually asked for.